PROBLEM DA12

Supersonic Missile Drag and Shock Standoff

Original statement · DA12.pdf

Π theorem | Mach similarity | Order of magnitude | Hypersonic limit

OFFICIAL PROBLEM STATEMENT

A nuclear missile is designed to fly at supersonic speeds (velocities U larger than the sound velocity c) at high altitudes. In the solution that appears, a bow shock is present at a standoff distance d from the missile tip. To characterize the solution and determine the drag force F acting on the missile, a series of experiments is conducted in a supersonic wind tunnel using a small experimental model geometrically similar to the real prototype. Analyze the results following these steps:

1. Use the Π theorem to reduce the parametric dependence of the drag F and standoff distance d. To set up the problem, note that the compressibility of the gas is characterized by the ambient sound speed c, which should be considered at the start of the analysis, along with the ambient values of the density and viscosity of the gas ρ and μ. Since the density changes across the shock, gravity might also influence the solution through the resulting buoyant force.

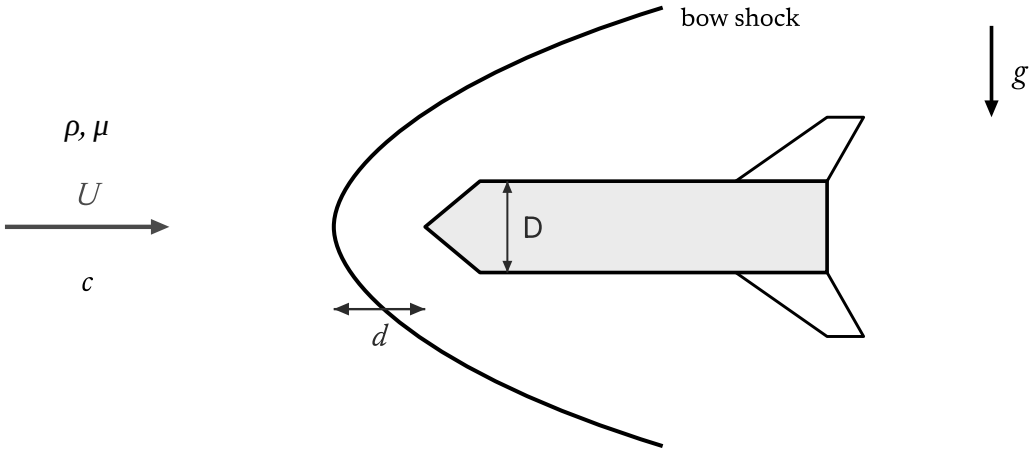
2. Derive the conditions under which both gravity and viscosity do not influence the solution. The reasoning should be substantiated with an order-of-magnitude analysis of the momentum equation to show clearly which dimensionless parameters measure the relative importance of buoyant and viscous forces (relative to the convective acceleration).

3. The experimental tests are carried out at normal ambient conditions ($\rho_e = 1.22$ kg/m^3, $\mu_e = 1.98 \times 10^{-5}$ kg/(m s), $c_e = 340$ m/s) with a model of diameter $D_e = 0.01$ m, significantly smaller than that of the real missile ($D_m = 1$ m), giving the results shown in the table. Use the results, together with the assumption that g and μ have a negligible influence, to deduce the variation with flight speed U_m of the drag force F_m and standoff distance d_m corresponding to the missile flying at cruise altitude, where $\rho_m = 0.36$ kg/m^3, $\mu_m = 1.44 \times 10^{-5}$ kg/(m s), and $c_m = 295$ m/s.

4. Check that the criteria needed to neglect g and μ are satisfied both in the experiment and for the real missile conditions.

5. Determine a simplified expression for the drag force when the Mach number is very large (i.e., $U/c \gg 1$).

U_e (m/s)	F_e (N)	d_e (m)
400	9.2	0.012
700	37.50	0.01
1,000	86.00	0.005
2,000	383	0
4,000	1532	0



Statement figure (redrawn from DA12.pdf, frame of the missile): a gas of density ρ, viscosity μ and sound speed c meets the missile of diameter D at a speed U > c; a bow shock stands at a distance d ahead of the tip

Source: DA12.pdf (page 1, prefinal exam of Dec 16, 2013, statement and handwritten key). Notes.pdf, Chapter 6: Π theorem, Eq. 6.25; order of magnitude, Eqs. 6.45-6.46; limits, Eq. 6.55; physical similarity, Eqs. 6.57-6.59.

Worked solution

Hypotheses, Data and Dimensional Table

- In the frame of the missile the gas arrives at U with its ambient ρ, μ, c; gravity is kept at first, as the statement asks, because the density jump across the shock creates a buoyant force.
- The model is geometrically similar; the subscript m is the missile (prototype), e the experiment.

Parameter	Dimensions
F (unknown)	MLT^{-2}
d (unknown)	L
D	L
U, c	LT^{-1}
ρ	ML^{-3}
μ	$ML^{-1}T^{-1}$

Parameter	Dimensions
g	LT^{-2}

Application of the Π Theorem

$$F = F(D, U, \rho, \mu, c, g), \qquad d = d(D, U, \rho, \mu, c, g)$$

$n = 6$; (ρ, U, D) have independent dimensions ($\det = -1$, as in DA1), so three groups remain.

- **Drag:** $[F] = [\rho][U]^2[D]^2$ (exponents as for the sphere, Notes.pdf, Eq. 6.32): $F/(\rho U^2 D^2)$.
- **Standoff:** d/D.
- **Viscosity:** $\rho U D/\mu$ (Reynolds). **Gravity:** $[g] = [U]^2[D]^{-1}$ (as in DA1): $U^2/(gD)$ (Froude squared). **Compressibility:** U/c (Mach).

$$\frac{F}{\rho U^2 D^2} = f_1\left(\frac{\rho U D}{\mu}, \frac{U^2}{gD}, \frac{U}{c}\right), \qquad \frac{d}{D} = f_2\left(\frac{\rho U D}{\mu}, \frac{U^2}{gD}, \frac{U}{c}\right)$$

Order of Magnitude: When Gravity and Viscosity Do Not Matter

Velocities are of order U and change by $O(U)$ over distances $O(D)$, as in Notes.pdf, Eq. 6.45. The convective acceleration, the viscous force and the buoyant force per unit volume are then

$$O(\rho\, \bar{v} \cdot \nabla \bar{v}) \sim \frac{\rho U^2}{D}, \qquad O(\mu \nabla^2 \bar{v}) \sim \frac{\mu U}{D^2}, \qquad O(\Delta\rho\, g) \sim \rho g$$

where the density change across a strong shock is of the order of ρ itself. The two ratios to the convective term are

$$\frac{O(\rho\,\bar{v}\cdot\nabla\bar{v})}{O(\mu\nabla^2\bar{v})}=\frac{\rho U^2/D}{\mu U/D^2}=\frac{\rho U D}{\mu},\qquad \frac{O(\rho\,\bar{v}\cdot\nabla\bar{v})}{O(\rho g)}=\frac{\rho U^2/D}{\rho g}=\frac{U^2}{gD}$$

So viscosity is negligible if $\rho U D/\mu \gg 1$, and gravity if $U^2/(gD) \gg 1$. Then (Notes.pdf, Eq. 6.55)

$$\frac{F}{\rho U^2 D^2}=f_1\left(\frac{U}{c}\right),\qquad \frac{d}{D}=f_2\left(\frac{U}{c}\right)$$

Mach Similarity: from the Tunnel to the Missile

Each tunnel point is similar to the flight state with the same Mach number:

$$\frac{U_m}{c_m}=\frac{U_e}{c_e}\implies U_m=\frac{c_m}{c_e}\,U_e=\frac{295}{340}\,U_e=0.8676\,U_e$$

and then

$$\frac{F_m}{\rho_m U_m^2 D_m^2}=\frac{F_e}{\rho_e U_e^2 D_e^2}\implies F_m=\frac{\rho_m}{\rho_e}\left(\frac{U_m}{U_e}\right)^2\left(\frac{D_m}{D_e}\right)^2 F_e=\frac{0.36}{1.22}\times 0.7528\times 10^4\,F_e=2221\,F_e$$

$$\frac{d_m}{D_m}=\frac{d_e}{D_e}\implies d_m=\frac{D_m}{D_e}\,d_e=100\,d_e$$

U_e **(m/s)**	U_m **(m/s)**	F_m **(N)**	d_m **(m)**
400	347	2.04×10^4	1.2
700	607	8.33×10^4	1.0
1000	868	1.91×10^5	0.5

U_e **(m/s)**	U_m **(m/s)**	F_m **(N)**	d_m **(m)**
2000	1735	8.51×10^5	0
4000	3471	3.40×10^6	0

Discrepancy with the official solution

The key writes $F_m = 1.91 \times 10^4$ N in the third row; $2221 \times 86 = 1.91 \times 10^5$ N, and the drag must grow between the second row (8.33×10^4 N) and the fourth (8.5×10^5 N), so the exponent of the key is a slip: the value is 1.91×10^5 N.

Check of the Criteria

The groups are smallest at the lowest speed of each series:

$$\left(\frac{\rho_e U_e D_e}{\mu_e}\right)_{min} = \frac{1.22 \times 400 \times 0.01}{1.98 \times 10^{-5}} = 2.46 \times 10^5 \gg 1, \qquad \left(\frac{\rho_m U_m D_m}{\mu_m}\right)_{min} = \frac{0.36 \times 347 \times 1}{1.44 \times 10^{-5}} = 8.68 \times 10^6 \gg 1$$

$$\left(\frac{U_e^2}{g D_e}\right)_{min} = \frac{400^2}{9.81 \times 0.01} = 1.63 \times 10^6 \gg 1, \qquad \left(\frac{U_m^2}{g D_m}\right)_{min} = \frac{347^2}{9.81 \times 1} = 1.23 \times 10^4 \gg 1$$

All four are very large: gravity and viscosity can be neglected in the tunnel and in flight.

Hypersonic Limit

For $U/c \gg 1$ the Mach number stops influencing the drag coefficient (Notes.pdf, Eq. 6.55):

$$\frac{F}{\rho U^2 D^2} = C$$

The tunnel data give C:

U_e/c_e	$F_e/(\rho_e U_e^2 D_e^2)$
1.18	$9.2/(1.22 \times 1.6 \times 10^5 \times 10^{-4}) = 0.471$
2.06	$37.5/(1.22 \times 4.9 \times 10^5 \times 10^{-4}) = 0.627$

U_e/c_e	$F_e/(\rho_e U_e^2 D_e^2)$
2.94	$86/(1.22\times10^{6}\times10^{-4}) = 0.705$
5.88	$383/(1.22\times4\times10^{6}\times10^{-4}) = 0.785$
11.8	$1532/(1.22\times1.6\times10^{7}\times10^{-4}) = 0.785$

The last two Mach numbers give the same value, so $C \simeq 0.785$ and

$$F \simeq 0.785\,\rho U^2 D^2 \qquad (U/c \gg 1)$$

Answers and Checks

- Reduced laws: $F/(\rho U^2 D^2)$ and d/D are functions of $\rho U D/\mu$, $U^2/(gD)$ and U/c.
- Viscosity and gravity are negligible for $\rho U D/\mu \gg 1$ and $U^2/(gD) \gg 1$; then both depend on U/c only.
- Missile: $U_m = 0.868\,U_e$, $F_m = 2221\,F_e$, $d_m = 100\,d_e$ (table above).
- Criteria: Reynolds $\geq 2.4\times10^{5}$ and Froude squared $\geq 1.2\times10^{4}$ in both systems.
- Hypersonic drag: $F \simeq 0.785\,\rho U^2 D^2$.
- Dimensions: $\rho U^2 D^2$ is $ML^{-3}\cdot L^2T^{-2}\cdot L^2 = MLT^{-2}$, a force.
- Physical sense: as the Mach number grows the shock moves closer to the tip ($d \to 0$ from $U_e = 2000$ m/s on) and the drag coefficient levels off, the behaviour predicted by (6.55).
- Check against the official key: DA12.pdf gives the same groups, $U_m = (c_m/c_e)U_e$, the same table (with the slip of the third force corrected above), the criteria 2.46×10^{5}, 8.67×10^{6}, 1.63×10^{6}, 1.23×10^{4} and $C \simeq 0.785$.